

Benchmark for Crack Re-Identification in Ticket Management Systems

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Abstract—In a ticket-based maintenance system, each new photograph of a crack should be assigned to the correct ticket, but if the same crack is photographed from a different angle it may be treated as a new crack, thus breaking its historical record and leading to redundant inspection work. We refer to this as crack re-identification (re-ID), which involves determining whether two photographs depict the same crack. This study presents CRACKID, a dataset consisting of 140 photographs of 17 walls and containing 212 distinct crack identities, and uses this dataset to assess ten zero-shot, crop-based matchers drawn from four different families under three scenarios: retrieval, thresholded pairwise decisions, and open-set identification with rejection. The ranking quality varies considerably (Rank-1 scores range from 0.370 to 0.613), although the validation-calibrated pairwise decisions only very slightly exceed the best constant baseline (F_1 scores of 0.331 to 0.338 compared to 0.305). Experiments involving context and crack erasure demonstrate that cues related to the wall's location are more important than the appearance of the crack. A geometric reference baseline that aligns the texture of crack-free wall sections and then compares the projected crack masks achieves a detection and identification rate (DIR) of 0.768 at a false alarm rate (FAR) of 0.1 on the 48% of the images that can be aligned, which is twice the DIR of 0.384 obtained by the best crop-based method. These point estimates are based on a small sample of walls, with each wall group being labelled by a single annotator using homography, so the advantage in the open-set setting depends on the source of the annotations, and the evaluation only considers correspondence within a single visit. These results indicate that good retrieval performance by itself does not guarantee reliable association of cracks and show the value of local visual context and geometric alignment in re-identification. By jointly evaluating retrieval, the matching decisions, and the rejection of unknown defects, CRACKID provides a foundation for developing more reliable visual association methods to support defect tracking in ticket management systems, with cross-visit evaluation and ticket-level duplicate reduction being future work.

Index Terms—Annotation Quality, Benchmark, Building Maintenance, Crack Re-Identification, Geometric Alignment, Homography, Structural Inspection

I. INTRODUCTION

Ticket management systems record reports of physical defects so that they can be inspected, prioritized, repaired, and followed up. Ideally, each ticket holds the history of a single incident. In practice, different users may photograph the same crack more than once, producing duplicate tickets that fragment this history and cause redundant assessment work. Every new observation must therefore either be linked to an existing ticket or open a new ticket for damage that has not yet been recorded.

This is a visual re-identification (re-ID) problem in which (i) the same physical crack must be recognised despite variations in viewpoint, camera distance, field of view, illumination, and partial visibility; (ii) a crack may itself change—by extending, widening, or branching—and must still be matched to its earlier observations, while genuinely new damage must be distinguished from growth of known damage. Errors in either direction lead to duplicate tickets or misattributed history.

Existing work on crack detection and segmentation focuses on single images. Datasets such as CrackSeg9k [1] and OmniCrack30k [2] contain no cross-observation identities, so observations cannot be associated with tickets. Whether techniques from person and object re-ID [3], [4] transfer to cracks is untested, since different defects may look alike while a single defect can vary considerably in appearance.

This work introduces CRACKID to evaluate whether photographs depict the same physical wall crack: 140 photographs of 17 walls captured during a single visit. Real photographs capture the acquisition changes that occur within a visit, whereas appearance and growth changes are probed only with synthetic images. Cross-visit evaluation on real photographs and ticket-level duplicate reduction are left for future work. All accuracy figures are point estimates from a small sample grouped by wall.

We assess ten zero-shot matchers using CRACKID. The protocol divides the process into two parts: the retrieval of candidate images and the decision as to whether two observations depict the same defect, and it includes open-set identification with rejection—that is, open-set association where the top match is accepted only if its score exceeds a threshold, otherwise the query is rejected as a new ticket—since a high rank by itself is not enough to associate a report with a ticket. To test whether matchers recognize the crack itself or only wall context, we use crack-erased control cases together with a geometric baseline which aligns the wall images and compares the projected crack masks.

Three research questions guide the study:

- RQ1.** Can the evaluated methods associate observations of the same defect under varying image acquisition conditions?
- RQ2.** How do wall context and geometric alignment contribute to defect re-identification?
- RQ3.** How do two selected approaches perform under synthetic changes in appearance and crack growth?

RQ1 and RQ2 are answered with real single-visit photographs; RQ3 is an exploratory study on synthetic data.

The contributions are as follows: (i) the cross-observation identities of CRACKID, together with their annotation provenance (Section III-B); (ii) a benchmark covering retrieval, same-defect decisions, and open-set identification, which addresses RQ1 (Sections IV and VI); (iii) context controls, a geometric reference, and a coverage analysis, which address RQ2 (Sections V, VI-C, and VII-B); and (iv) a synthetic revisit probe, which addresses RQ3 (Section VII-C).

II. RELATED WORK

Most crack-analysis research focuses on detecting or segmenting cracks in individual images [5]–[7]. Consolidated datasets [1], [2] retain this image-level formulation and do not provide identity annotations across observations. Although multi-view pipelines establish geometric correspondences to fuse images acquired during a single inspection into surface models or aggregated segmentations [8], they do not explicitly represent persistent crack identities across observations. Structural-health monitoring tracks cracks over time using fixed cameras and designated locations [9]–[12], where crack identities are *assigned at construction time*; unlike our open-set association task, they track known cracks.

The scenario involves object re-ID [4] with open-set identification [13]: Scale-Invariant Feature Transform (SIFT) [14] and Oriented FAST and Rotated BRIEF (ORB) [15], the learned correspondences such as SuperGlue [16] and LoFTR [17], and the region embeddings [3], [18]–[21] are all assessed in a zero-shot manner without CRACKID fine-tuning (the YOLOv8 row: layer-9 `yolov8n.pt` features [22]). LightGlue [23] and NetVLAD [24] are mentioned as future retrieval baselines. Vehicle-damage re-ID across claims [25] follows the same open-set protocol as ours; related work focuses on detection and severity, not on association [26]–[28]. The approach of aligning first and then reasoning about presence or absence constitutes change detection [29], [30] using planar registration [31]–[33] and an Enhanced Correlation Coefficient (ECC) fallback [34]; our key point is that the alignment and the matching targets must be *disjoint*, with the homography being derived from the wall when the crack is identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID dataset consists of 140 photographs of 17 exterior walls that were taken at a single site over four half-days using two consumer handheld devices (Mendeley Data [35]; Fig. 1). The walls are either stippled render (textured) or smooth painted plaster, and the only features that are visible are cracks. The dataset is divided such that six walls are used for validation (walls 01 to 06; 66 photos, 66 identities) and eleven walls for testing (walls 07 to 17; 74 photos, 146 identities). Only those identities that appear in at least two photographs are included in the answerable queries (18 validation, 19 test; $n_Q=439$ validation and 305 test queries). We added Walls 18 and 19 to the dataset afterward

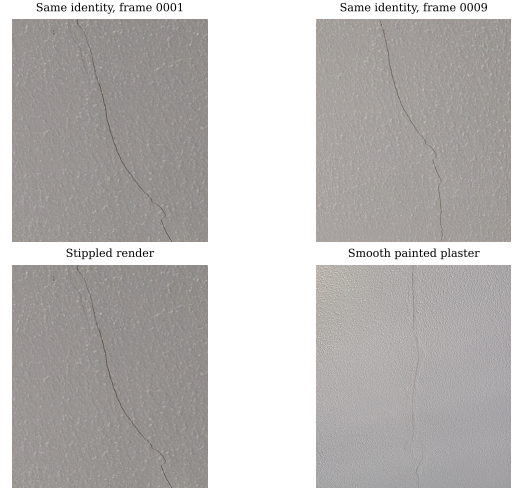


Fig. 1. Real CRACKID examples. Top: one identity across non-adjacent frames. Bottom: stippled render versus smooth plaster. Labels via homography propagation, unaudited.

and therefore exclude them from all results. Each wall was photographed during a continuous walk-around (median time of 3 seconds between frames and a maximum of 81 seconds); the data include within-visit viewpoint and scale correspondences, providing a more favorable scenario than multi-visit matching, not a formal upper bound.

For adjacent-frame retrieval, the gallery includes the adjacent frame for all 305 test queries that require an answer. The gap sweeps described in Sec. VII-B (involving 305 queries down to 205, then to 149, and finally to 105 for gaps 0–3) maintain Rank-1 as the number of items in the pool decreases, which shows that the method is robust when faced with short baselines.

B. Annotation, and why it needs auditing

The ORB homography method carries every single click from frame to frame and then combines the fragments to form physical-crack identities; the merging process is the greatest risk (since an incorrect merge can turn a hard negative into a false positive). In the frozen snapshot, 95 of the 212 identities (45%; 70 of 146, 48% in the test case) result from merges, with median bridged gaps of 228 pixels (test) and 286 pixels (validation). One annotator examined all the clicks and sorted the unclaimed points (107 of the 1,023 were classified as no crack; the details are archived with the code). Although single-annotator triage reduces the merge risk, it does not eliminate it. All the figures refer to the frozen snapshot (Table I); the live dataset/labels/ directory includes walls 18–19 and must not be combined with the frozen figures without first re-running the benchmark.

IV. BENCHMARK PROTOCOL

The benchmark assesses visual association by determining, for a new crack observation, whether it matches one of the recorded defects; since annotated identities serve as a stand-in for permanent defect records, the scores indicate the accuracy

TABLE I
COUNTING UNITS PER METRIC (FROZEN SNAPSHOT, $n_Q=305$ TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	305
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

of the association rather than the reduction of duplicates at the ticket level.

A. Task formulation and candidate records

There is one physical wall; an identity connects the different observations of the same crack (Section III-B). A query refers to one instance of a crack; its gallery includes other instances taken from photos of the same known wall. Cross-wall search is not allowed. A query is called *known* if its gallery contains the identity, otherwise it is *unknown* (meaning it is a genuinely new defect). In closed-set retrieval it is assumed that the answer will always be in the gallery and the candidates are only ranked. Open-set identification, on the other hand, involves modelling the association of tickets: the system has to either link the query to the best existing ticket or reject it and create a new ticket if no gallery entry scores high enough. Each method evaluates query-gallery pairs for the purposes of ranking, makes thresholded decisions and carries out identification with rejection. Pairs are the unit for pairwise decisions; queries are the unit for retrieval and for open-set decisions; image pairs are the unit for registration and assignment. The Hungarian one-to-one constraint is applied only for each image pair.

Counts for the frozen snapshot (6 validation walls 01–06; 11 test walls 07–17; walls 18–19 excluded) are in Table I. Registration covers all 301 same-wall test image pairs; 255 yield scored pairs with validity-mask cells (Sec. VII-B).

B. Test conditions, decisions, and metrics

The primary test involves actual walk-arounds for each wall, assuming that the morphology remains unaltered; the only variations considered are viewpoint, distance, scale, lighting, and visibility, along with same-wall distractors and unknown queries. The validation and test walls are entirely separate. All the methods are used exactly as they are on the shared masks without any tuning of CRACKID (see Sec. V); the thresholds determined on the validation set are applied to the test set, considering only within-visit correspondence. A synthetic check is carried out to distinguish between photometric effects (condition B: blur/noise/contrast with unchanged structure) and growth effects (condition A: one tip extended) using ten photos taken from five walls with a single seed and a common warp (the parameters are available with the code), and this is done by comparing OSNet with context to HYBRID-SKEL

($\tau=8.0$ px), which is different from REG+CHAMFER: this is only a preliminary check.

Decisions: any pairs that exceed the validation-calibrated threshold are accepted; for open-set identification, only the highest-ranking gallery entry above the threshold is accepted for each query, and a query with nothing above threshold is rejected as a new ticket. For each image block, the Hungarian assignment is used with those below the threshold being dropped. Unscorable pairs are retained (they are predicted to be negative for F_1 and are given the last rank in retrieval); a registration failure is defined as the absence of evidence, not as evidence of a new defect. The metrics used are Rank-1 and mean Average Precision (mAP) for closed-set ranking; pairwise F_1 compared with the oracle F_1^* versus the validation $F_1^{@v}$; assignment F_1^a equal to $2TP/(2TP+FP+FN)$; and, for open-set, detection and identification rate (DIR) at false alarm rate (FAR) of 0.1—that is, the fraction of known queries correctly linked when at most 10% of unknown queries are wrongly linked—together with coverage (the fraction that is scored). The references include random ranking and the best-constant F_1 $2p/(1+p)$ (0.305 at a prevalence of 0.180; 0.544 on the 0.48 geometric subset; 0.452 in the validation at 0.292); the full chance levels are given in Table II. Since there are 11 walls and 19 matchable identities, we provide the point estimates (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack locations on the wall drive the association: it aligns wall surfaces to a common frame and then compares the projected crack masks. It assumes a roughly planar wall and uses no CRACKID training. The shared masks (trained on public crack data only; no CRACKID wall in training) use instance extraction at 200 px and 5 px to keep inputs consistent across methods. Masks are taken as given, and their error is not measured.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from the surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and the succeeding stage, not guaranteed per pair. Full detector settings (Scale-Invariant Feature Transform (SIFT) with Contrast Limited Adaptive Histogram Equalization (CLAHE) [36], ratio test, robust estimation) are archived with the code.

B. Registration cascade and validity checks

Textured render generally provides background features, whereas smooth plaster does not. A three-stage cascade is used: masked SIFT; if too few keypoints remain, unmasked SIFT as fallback (flagged per pair, since the crack then aids alignment); finally, Enhanced Correlation Coefficient (ECC) photometric alignment. Candidates pass plausibility tests (no mirrored or folded quads; scale within $[1/16, 16]$); the surviving stage is recorded per pair, and its inlier root-mean-square

error (RMSE) sets the comparison tolerance. Unscored pairs count as lack of evidence, not evidence of different defects.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance in (1), which tolerates the small misalignments that collapse intersection over union (IoU) on narrow cracks. Let a, b be crack point sets and $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right]. \quad (1)$$

Early rejection of distant candidates applies. Chamfer coverage (fraction within tolerance τ of the per-pair RMSE, clamped to $[2, 16]$ pixels, default 6 pixels; HYBRID-SKEL uses 8.0 pixels) is evaluated separately as its scale-free counterpart (see Sec. VII-B).

D. Correspondence decisions and limits of spatial evidence

For each image pair, one-to-one assignment follows the Hungarian method [37] with the validation threshold shared by all methods; unscored pairs take value $-\infty$ (predicted negative, ranked last, as in Sec. IV). An aligned projection onto visibly empty wall is a *licensed negative* (evidence of absence, stronger than low embedding similarity). At the same time, missed detections, out-of-frame areas, and registration failures stay unresolved. The REG+CHAMFER reference targets unchanged damage; progression uses HYBRID-SKEL. Performance depends on surface-dependent failures and on homography-based annotation provenance.

VI. THE APPEARANCE CEILING

A. Setup

The following ten methods employ crack-centered crops together with a common validity mask: SIFT and ORB combined with Random Sample Consensus (RANSAC) inliers, SuperGlue and LoFTR using pre-trained indoor weights, five embedding methods, and re-ID OSNet with cosine similarity, all of which are applied in a zero-shot manner on CRACKID; YOLOv8 makes use of the layer-9 `yolo8n.pt` features. The input sizes and checkpoints are available in the code. For context control, fixed backbones (OSNet, CLIP) are used: `@ctx1` increases the size of the instance box by $1 \times$ in each direction; `@ctx1@nocrack` fills in the crack in order to isolate the wall context. REG+CHAMFER first aligns the walls and then compares the projected masks (as described in Sec. V); COVERAGE-ONLY continuously evaluates the registered pairs without examining the cracks. The synthetic tests compare OSNet with context against the HYBRID-SKEL variants.

Timing is wall-clock on a Google Colab NVIDIA T4 GPU runtime with 12.7 GB system RAM and 15.0 GB GPU RAM (CUDA when available, otherwise CPU), measured with `perf_counter` as `prepare + score` over the full test scoring pass with no batch-size tuning: embedding matchers forward all crops in one batch (YOLOv8 loops per crop), keypoint

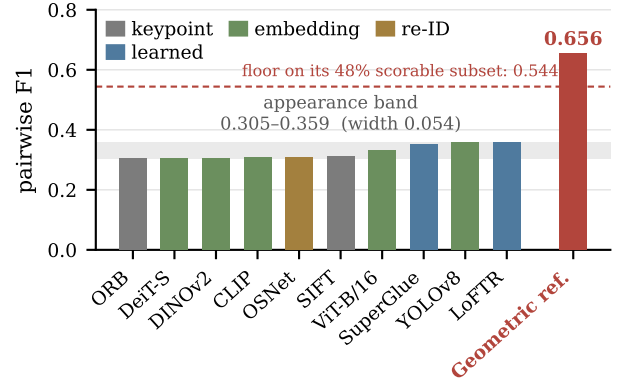


Fig. 2. Appearance ceiling: pairwise F_1^* stays narrow while ranking spreads. Geometry (red) leaves the band but uses the 0.48 pool with floor 0.544 (Table II).

and learned matchers score per valid query-gallery pair, and geometry registers once per same-wall image pair (301 test pairs, pruned to scorable cells). We report `s/query` as total time divided by test queries and, for geometry only, also `s/pair` as total time divided by image pairs. Eight crop methods plus controls ran together; SuperGlue and LoFTR were re-run separately to add validation thresholds, so costs are not load-controlled and are relative only.

B. Result: appearance buys almost nothing on the decision

Table II (11 walls, frozen snapshot, $n_Q=305$) displays a Rank-1 score ranging from 0.370 to 0.613 (with the best result being that of ViT-B/16; the highest mAP is achieved by OSNet at 0.497; and the best crop performance with DIR at a FAR of 0.1 is 0.384 for SuperGlue), although the oracle F_1^* lies between 0.337 and 0.374 and the validation-calibrated $F_1^{@v}$ is in the range 0.331 to 0.338: therefore, a high ranking does not mean that decisions have been made. The small range of the $F_1^{@v}$ values indicates a poor transfer of thresholds across the walls. Since there are 19 matchable test identities, we do not claim to have any fine-grained crop ranking. All the values given are point estimates.

C. Is it the crack, or the wall behind it?

Context contributes to both ranking and open-set performance, with most of this gain retained when the crack is erased: for OSNet the Rank-1 score rises from 0.603 to 0.775 (falling to 0.711 when the crack is erased), the DIR at FAR=0.1 goes from 0.154 to 0.404 (down to 0.393 when erased); for CLIP it goes from 0.143 to 0.311 (down to 0.286 when erased). The erased views outperform all the tight crops, suggesting wall context is the cue. The pairwise F_1^* remains between 0.307 and 0.361 for the base, context, and erased views, although open-set performance improves by +0.250 for OSNet. Within-visit retrieval does work, but reliable automatic decisions do not; ranked lists might help the operators, though more evidence is needed regarding acceptance (Fig. 2).

TABLE II

MAIN RESULT ON CRACKID TEST (11 WALLS, FROZEN SNAPSHOT, $n_Q=305$). F_1^* USES AN ORACLE TEST THRESHOLD; ΔF_1^* IS EXCESS OVER THE FULL-POOL FLOOR; $F_1^{\text{@v}}$ USES THE VALIDATION THRESHOLD. BOLD IS THE BEST CROP PER COLUMN. GEOMETRY COVERS 0.48 WITH FLOOR 0.544, SO ITS ΔF_1^* IS NOT COMPARABLE TO CROP ROWS (SEC. VII-B). S/QUERY IS TOTAL WALL-CLOCK (PREPARE + SCORE) PER TEST QUERY ON A COLAB T4 GPU; RELATIVE TIMES ONLY FROM TWO SESSIONS (SEC. VI-A).

Method	Family	R@1	mAP	DIR @FAR= 0.1	F_1^*	ΔF_1^*	$F_1^{\text{@v}}$	cov.	s/q
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [22]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometry attains a Rank-1 score of 0.952 as compared to the best crop result of 0.613 (ViT-B/16) and an mAP of 0.886 compared to 0.497 (OSNet), with a coverage of 0.48 rather than 1.00. After coverage has been exceeded, ranking improvements continue but open-set gains do not: at a FAR of 0.1, DIR is 0.768 as against 0.384 (SuperGlue), 0.404 (the best OSNet+ctx1), and 0.000 for coverage (the chance level being 0.038). In the case of hairline cracks, crop similarities almost never reject, and aligned empty projections function as valid negatives. For cost (timing method in Sec. VI-A, relative only), LoFTR takes 7.19 s/query versus 0.006 s/query for OSNet, while geometry takes 27.9 s/query and 26.0 s/pair. With regard to RQ2, alignment allows identification beyond coverage, on the condition that we take into account success, surface, and provenance. The point estimates are based on 19 identities.

B. Coverage, failures, and ablations

Crops achieve 100% of the pairs while geometry reaches 48%. Unsourced pairs receive a negative score and are assigned the lowest rank. Censoring reduces ranking accuracy. Still, it increases the pairwise F_1 : the scored sets retain 100% of the positives but only 36.8% of the negatives (prevalence ranging from 0.180 to 0.373; floor ranging from 0.305 to 0.544). The coverage-only approach yields an F_1 of 0.544, a mAP of 0.533, and a Rank-1 of 0.515, so almost all of the ranking and pairwise performance comes down to *which pairs align*; the only exception is the open-set lead (0.768 versus 0.000). The failures depend on the wall (zero on two test walls; a per-wall table is provided with the code). Adjacent frames answer all 305 queries, so the ranking is based on a short baseline; the gap sweeps (going from 305 down to 105 at a gap of 3) maintain

TABLE III

SYNTHETIC REVISIT SCENARIO: HYBRID-SKEL VERSUS OSNET+CTX, WITH THE F_1 SCORE AT THE VALIDATION THRESHOLD; ONE SEED AND FIVE WALLS. HYBRID-SKEL DIFFERS FROM REG+CHAMFER AS IN TABLE II.

Protocol	Method	n_q	R@1	mAP	DIR @FAR _{0.1}	$F_1^{\text{@v}}$	F_1^a
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.631
B (appear.)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609

Rank-1. Registration is determined by sharpness rather than by frame gap (Wald $z = 5.11$, $p = 3.3 \times 10^{-7}$ versus $z = -2.78$, $p = 5.4 \times 10^{-3}$); with sharpness at least 10, registration is achieved in 56.1% of the cases when one wall is removed from the filtered set. Scale-free coverage results in a loss of 0.240 mAP and 0.103 Rank-1 when compared to Chamfer (DIR increased by +0.014). When crack exclusion is disabled, the ranking remains unchanged, but DIR increases by +0.057; masking is retained for independence. ECC contributes approximately 0.03 to Rank-1 and 0.02 to mAP. Geometry totals 7817.3 s over 301 image pairs (26.0 s/pair; 27.9 s/query in Table II), about three times LoFTR and more than 100 times the cheapest embedding (relative, Colab T4; Sec. VI-A).

C. A synthetic revisit protocol

Ten source photos taken from five walls (with one seed) are subjected to the same minor warp in either case (A) under single-tip growth or (B) due to photometric corruption, and then are matched with the actual images (procedure with code; p90 tilt 75°). HYBRID-SKEL differs from REG+CHAMFER as shown in Table II. Table III compares HYBRID-SKEL and OSNet+ctx.

For ranking/open-set, OSNet+ctx outperforms HYBRID-SKEL (R@1 0.826/0.671 versus 0.275/0.274 and DIR 0.739/0.663 versus 0.343/0.342), while on assignment- F_1^a the order reverses narrowly (0.646/0.640 versus 0.631/0.609; $F_1^{@v}$ remains similar at 0.569/0.574 versus 0.601/0.594); no cell corresponds to the pure homography case ($H = 0.0\%$). For RQ3, this single-seed synthetic probe (Table III; ten source photos, five walls) provides no evidence for real multi-visit robustness, which remains untested (Sec. VIII).

VIII. LIMITATIONS AND FUTURE WORK

This study presents the first measurable test of re-identifying a crack in a single visit using a geometric reference. The results are derived from a small number of clustered walls, labels provided by a single annotator, and partial registration coverage. Each of the points listed here corresponds to a finding together with a step for verifying it.

- 1) **Scale to more walls and visits.** Collect data from more walls, different surface types, and additional annotators to improve the confidence intervals and the reliability of the rankings. Most importantly, include a second session using different distances and lighting conditions to establish a genuine cross-visit benchmark.
- 2) **Add provenance-free annotation.** One annotator reviewed all clicks and handled the unclaimed points; agreement was not assessed, and the geometric advantage depends on provenance. To detect the gap independent of provenance, it would help to create a small set of labels without homography and share score matrices per pair for sensitivity to wall-clustered intervals and threshold-free AP/AUROC.
- 3) **Close the coverage gap.** The baseline covers 48% of the pairs (97 out of 301 image pairs; 15,328 out of 31,822 valid cells) and is ineffective when dealing with smooth plaster. By using surface-adaptive registration methods such as learned local features, direct photometric alignment, or sparse depth obtained from consumer devices, a more thorough evaluation could be achieved, and the separate geometry could be better distinguished from the homography-based provenance.
- 4) **Learn from geometric supervision.** The baseline is an untrained reference and not a system that has been put into operation. The location benefit could be obtained by using pseudo-labels from the registration process or by fine-tuning a foundation model on homography correspondences, without per-pair registration.
- 5) **Assess the ticket lifecycle.** Define task metrics such as the duplicate rate and the missed-new-defect rate, and also test segmentation, re-identification, and joint ticket assignment in a closed-loop simulation of real inspections.

Together, these steps will turn the within-visit reference into a multi-visit re-identification component that can be used operationally for tickets.

IX. CONCLUSION

We present CRACKID, which consists of 140 photographs from 17 walls, along with a protocol for retrieval, deciding whether defects are the same, and performing open-set identification with rejection.

The main findings are constructive: concerning (RQ1), ten zero-shot crop matchers rank above chance, which establishes a working retrieval baseline, while thresholded decisions remain close to the best-constant baseline and motivate joint evaluation of ranking, decisions, and rejection; concerning (RQ2), local wall context contributes substantially to appearance-based association, and the geometric reference demonstrates how surface alignment and projected crack overlap can support open-set identification on the 48% alignable subset; concerning (RQ3), the small synthetic revisit probe clarifies the boundary between within-visit association and multi-visit robustness.

These findings show that retrieval performance alone does not ensure reliable crack association and highlight the contribution of local visual context and geometric alignment to re-identification. By jointly evaluating retrieval, matching decisions, and rejection of unknown defects, CRACKID provides a foundation for developing more reliable visual association methods to support defect tracking in ticket management systems.

Because all figures are point estimates from a small sample grouped by wall, and a single annotator assigned the labels using homography, the open-set advantage depends on provenance, and the ranking measures consider only within-visit correspondence. This dataset and benchmark focus on the visual association stage. Further work is needed to evaluate across visits and reduce ticket-level duplicates.

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